FGR-GA: Integrating Fine-Grained Representation Refinement and Group-Aware Alignment for Remote Sensing Image–Text Retrieval

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***Abstract*—Leveraging powerful semantic understanding and generation capabilities, vision-language pretrained (VLP) large models have demonstrated remarkable potential in cross-modal retrieval. However, remote sensing scenarios present unique challenges: images are characterized by tiny objects and substantial background information or noise, while their corresponding textual descriptions are often concise and generic, hindering the accurate extraction of fine-grained modal representations. Additionally, existing VLP-based methods primarily adopt instance-level feature alignment, which focuses on reducing the distance between matched image–text pairs without meticulously accounting for global modal correlations. To tackle these issues, we propose FGR-GA, a novel VLP-based framework for remote sensing image–text retrieval, designed to improve cross-modal retrieval accuracy while ensuring computational efficiency. Specifically, building on initial modal features from a frozen multimodal large language model, we introduce a lightweight image feature disentanglement module (IFDM) and a key textural term annotation mechanism, aimed at refining fine-grained modality representations for images and text, respectively. Meanwhile, we develop a group-aware image–text similarity scoring function to enable modality alignment from a macro perspective, paired with a multiview contrastive learning strategy that optimizes the distribution of shared representations. This approach not only guides effective optimization of the shared space but also strengthens the model's ability to distinguish subtle semantic differences between samples, ultimately contributing to enhanced retrieval accuracy. Comprehensive experiments on the RSICD and RSITMD datasets have validated the superiority of our method over state-of-the-art baselines. Notably, by incorporating FGR-GA into fine-tuned CLIP, we obtain mR improvements of 1.92% and 1.59% with only 3.0% and 2.9% extra parameters.**

***Index Terms*—Cross-modal retrieval, modality alignment, remote sensing images, visual-language LLMs.**

I. Introduction

With the significant advancements in Earth observation technologies [1], [2], [3], [4], [5], [6], remote sensing image data has experienced explosive growth, providing massive and diverse data for geographic information analysis, environmental monitoring, natural disaster assessment, and urban planning [7], [8]. Nevertheless, the high dimensionality, inherent complexity, and diversity of remote sensing data pose severe challenges to feature extraction and content comprehension. For instance, remote sensing images are characterized by small objects, intricate backgrounds, and significant noise interference, while their textual descriptions tend to be broad and abstract. Furthermore, issues such as semantic imbalance and substantial distribution discrepancies in multimodal data further exacerbate the difficulty of precise image–text retrieval.

Existing remote sensing image–text retrieval methods can be categorized into two types: deep learning-based methods [12], [13], [14], [15], [16], [17], [18], [19], [21], [22], [23], [24], [25], [26] and those using visual-language pretrained (VLP) models [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44]. The former primarily extract image features via backbone networks [e.g., convolutional neural networks (CNNs) or vision transformers (ViTs [11])]. They then design dedicated network architectures and cross-modal alignment mechanisms to learn a shared image–text representation space for facilitating cross-modal matching. Their performance is highly dependent on the scale and quality of training dataset. Additionally, full retraining of all modules (including multimodal feature extraction and representation alignment) is required, leading to expensive computational and training costs. In contrast, VLPs (e.g., CLIP [10] and ALIGN [27]) can acquire universal and rich semantic representations through joint pretraining on large-scale cross-modal datasets, effectively narrowing the distribution gap between textual and visual features. Furthermore, in practical applications, these methods often adopt parameter-efficient fine-tuning strategies, such as updating a small subset of network parameters or freezing the model to serve solely as an initial feature extractor, thus significantly reducing computational overhead.

Despite notable advancements, VLP-based remote sensing image–text retrieval still confronts three key challenges.

1) *CH1. How to accurately extract fine-grained multimodal representations?* Remote sensing images contain small objects and highly similar backgrounds within the same scene, which impedes the discrimination of subtle intersample differences. Meanwhile, textual descriptions typically consist of sparse key semantic terms, making critical granular information (e.g., object attributes) overwhelmed.

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2) *CH2. How to improve fine-grained feature alignment without complex model architectures?* Existing VLP-based methods prioritize global semantic learning but lack fine-grained image–text feature alignment. Despite some studies attempt to boost performance by increasing network depth or scaling up pretrained models [10], experimental results indicate that higher model complexity does not effectively improve cross-modal retrieval accuracy.
3) *CH3. How to enhance the model's ability to distinguish minor semantic differences?* Current approaches primarily train models by minimizing the distance between matched image–text pairs, overemphasizing instance-level alignment while neglecting global modal correlations and shared representation optimization. This impairs the model's semantic discriminative ability, ultimately undermining retrieval accuracy (e.g., Recall@1 and Recall@5 are often significantly lower than Recall@10).

In this article, we introduce FGR-GA, a lightweight remote sensing image–text retrieval framework based on VLPs. It comprises four core modules: VLP-based feature extraction, fine-grained representation enhancement, image–text similarity scoring function design, and multiview contrastive learning strategy. *To address CH1*, we propose an effective fine-grained modal representation refinement method to reduce irrelevant information interference and emphasize key semantics. For images, a feature disentanglement module is constructed to efficiently separate background and object features in each patch without introducing complex network structures. Meanwhile, graph learning is performed to strengthen semantic associations between objects, laying a solid foundation for subsequent cross-modal matching. For text, we employ a key term annotation mechanism to identify core vocabulary strongly associated with remote sensing objects, and then highlight key semantic information in textual descriptions.

For CH2 and CH3, we propose a two-step solution to enhance cross-modal alignment and semantic discrimination. First, we design a novel image–text similarity scoring function that breaks the traditional "individual-to-individual" matching paradigm, enabling joint optimization of global semantic matching and group-aware fine-grained modal alignment. Taking text-to-image retrieval as an example: we first extract key semantics from the query text and select semantically relevant images from the training batch to form a reference group. The scoring function is then constructed by integrating two complementary terms: individual-level similarity between the query text and matched image, and macro-level distance between the query text and the "semantically related image group". Second, we propose a multiview contrastive learning strategy for model optimization. Unlike traditional instance-level contrastive learning, it no longer solely relies on direct pairwise sample associations, but takes the previously identified "semantically related samples" as reference anchors, while optimizing the shared representation space with triplet ranking loss between global image–text representations. The core principle is: highly similar samples are constrained to maintain small distances from the reference group; for dissimilar samples, the distance gap to the reference group is amplified to promote separation in the representation space. Through dual "individual-group" and "individual-individual" constraints, our method efficiently optimizes the shared representation space and enhances semantic discrimination, ultimately improving the accuracy of remote sensing image–text retrieval. Our key contributions are summarized as follows.

1) We propose a novel VLP-based framework for remote sensing image–text retrieval, which achieves effective and cost-efficient retrieval accuracy improvement through fine-grained representation refinement and group-aware cross-modal alignment.
2) To effectively extract fine-grained modality representations, we construct an image feature disentanglement enhancement module and a key textural term annotation mechanism for images and text, respectively, mitigating interference from irrelevant information and emphasizing the influence of key semantic elements.
3) We design a group-aware image–text similarity scoring function coupled with a multiview contrastive learning strategy. By considering global modal correlations, the proposed method jointly optimize fine-grained cross-modal alignment and shared representation space, strengthening the model's ability to distinguish nuanced semantic differences.

Extensive experiments on two benchmark datasets have validated the efficacy of our FGR-GA method. Notably, by constructing FGR-GA upon the pretrained CLIP model and GLISA (a SOTA baseline), we achieve an average accuracy improvement of approximately 5% while introducing merely 3% additional model parameters. This renders FGR-GA a promising versatile framework for adapting existing dual-tower cross-modal retrieval paradigms. Furthermore, ablation studies and visualization analyses have confirmed that FGR-GA effectively enhances the model's discriminative capability.

II. Related Work

A. Deep Learning-Based Remote Sensing Image–Text Retrieval

Conventional remote sensing image–text retrieval methods rely on unimodal feature extraction and cross-modal matching, which can be categorized into CNN-based and vision transformer (ViT)-based approaches by their backbone networks. For CNN-based methods, high-dimensional image features are extracted via hierarchical convolution and integrated with classifiers or matching mechanisms to achieve cross-modal alignment. Representative works include AMFMN [12] (using multiscale visual attention mechanism and dynamic variable-margin loss), GaLR [13] (adopting global–local semantic alignment), HVSA [14] (proposing hypersphere-curriculum learning), MGRM [15] (employing masks and transformers to enhance entity and relation modeling), and SCAT-PRG [16] (presenting spatial-channel attention with pseudo regions). Additional methods use multilayer outputs for capturing multiscale/fine-grained semantics [17], [18], [19], [20], while KAMCL [21] (knowledge-aided contrastive learning), MSITA

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[22] (multiscale learning), and DOVE [23] (directional embedding) are also notable. While these CNN-based methods perform well in capturing spatial structural information, they are limited in modeling long-range dependencies and global semantics.

ViT-based methods utilize self-attention mechanisms to model global image structures, enabling capture of higher level semantics and finer grained feature alignment. For instance, PIR [24] integrates prior knowledge and instructions to enhance representation alignment, thereby boosting retrieval performance; SMLGN [25] proposes a symmetric multilevel guidance network that fuses multisubspace learning with hierarchical guidance for remote sensing image–text matching; Ouyang et al. [26] introduce the sparse information perception network (SIPN), an instance-level dynamic model addressing the sparsity of remote sensing images. However, these methods typically require backbone retraining to adapt to specific remote sensing data distributions, incurring extra computational costs. Furthermore, existing fine-grained approaches often overemphasize all potential details, which may obscure key features and degrade retrieval accuracy.

B. VLP-Based Remote Sensing Image–Text Retrieval

Recent advances in VLP-based methods have significantly promoted cross-modal matching via joint pretraining of vision-language models on large-scale image–text datasets [10], [27], [28], [29], [30], [31], [32]. These approaches learn general semantic representations from massive data, enabling more efficient and accurate image–text retrieval. Specifically, some methods leverage VLP models (e.g., CLIP [10], ViLT [33]) as backbones for feature extraction to strengthen the capture of low-level semantics. For instance, IEFT [34] enhances feature interaction through transformers; MIIA [35] adopts a CLIP-based dynamic contrastive learning strategy to improve image–text alignment. Additionally, CLGSA [36] achieves precise semantic alignment in complex remote sensing scenes via local key feature alignment and cross-sample global semantic alignment modules; FGVLA [37] combines coarse-grained contrastive losses and fine-grained contrastive losses, boosting semantic alignment via explicit image-patch-to-text-noun mapping. CoCMN [38] filters intramodal redundant information and captures intermodal co-occurrence relations using a text–image attention module and a co-occurrent features module. Furthermore, FSSN [39] enhances feature robustness by jointly modeling frequency and spatial features.

To better leverage pretrained models, researchers have proposed methods targeting reduced training overhead and improving overall performance. Specifically, Yuan et al. [40] developed a parameter-efficient transfer learning (PETL) framework that combines CLIP with a multimodal remote sensing adapter and a hybrid multimodal contrastive learning objective. Ning et al. [41] proposed a representation discrepancy bridging (RDB) method, addressing imbalanced cross-modal optimization via a cross-modal asymmetric adapter for modality-specific feature enhancement and a dual-task consistency loss for robust alignment. Liu et al. [42] presented RemoteCLIP, a CLIP-based method with domain-specific adaptation for remote sensing data, and integrated a cross-pattern retrieval target to refine image–text pattern alignment. Hu et al. [43] proposed GLISA, a dual-level alignment framework that captures both global and local features from remote sensing data, improving cross-modal retrieval accuracy through soft alignment. Zhao et al. [44] proposed sparse-guided partial dense (SGPD) cross-modal retrieval, a plug-and-play method that converts dense vectors into sparse representations to enable efficient and accurate retrieval performance.

Notably, both conventional and VLP-based methods primarily rely on discriminative loss functions trained on individual instances, while overlooking common background noise among similar instances. Consequently, these methods often fail to account for intramodal data distributions, making it challenging to capture critical semantics and distinguish minor differences between samples.

III. Methodology

A. Framework Overview

In this article, we propose FGR-GA, a lightweight VLP-based framework for remote sensing image–text retrieval. Its aims to bridge the cross-modal semantic gap by enhancing fine-grained feature extraction, improving cross-modal alignment paradigm, and optimizing the shared representation space from both instance and macro levels. As shown in Fig. 1, our FGR-GA framework contains four core modules: initial feature extraction, fine-grained multimodal representation enhancement, group-aware image–text similarity scoring function design, and multiview contrastive learning.

First, we adopt CLIP as the backbone to extract initial multimodal representations. Specifically, global semantic embeddings are extracted for complete images and text, while fine-grained local embeddings are generated for image patches and text tokens, respectively. These embeddings are then refined via modality-specific enhancement modules to suppress noise and emphasize key information. For images, we design an image feature disentanglement module (IFDM) to separate background interference from object-related features in each image patch. Additionally, we leverage graph learning to strengthen interobject semantic correlations. For text, a key term highlighting mechanism is used to emphasize critical semantics, effectively mitigating interference from general or redundant expressions. Based on the enhanced fine-grained representations, we develop a novel group-aware image–text similarity function and a multiview contrastive learning strategy. Specifically, semantically relevant image groups are used as reference anchors to decompose text semantics and calculate contrastive loss from the instance-level and group-level perspectives. By optimizing the shared representation distribution, we effectively improve the accuracy of remote sensing image–text retrieval. Next, we elaborate on the implementation of each module.

B. Initial Feature Extraction

Consider a dataset containing M image–text pairs, denoted as $\mathcal{M} = \{(I^{(m)}, T^{(m)}, \boldsymbol{y}^{(m)})\}_{m=1}^{M}$, where $I^{(m)} \in \mathbb{R}^{H\times W\times C}$ represents the mth remote sensing image with a spatial size of

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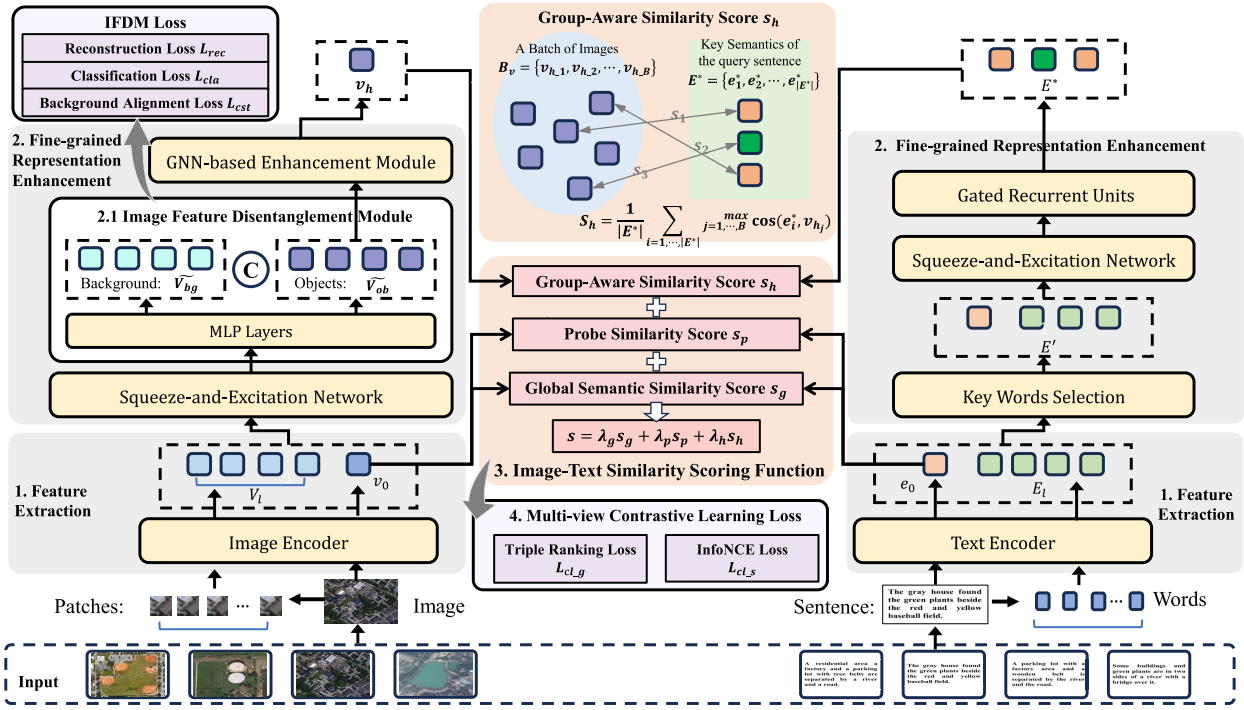


Fig. 1. Overview of our FGR-GA framework. It comprises four core modules: initial feature extraction, fine-grained multimodal representation enhancement, group-aware image–text similarity scoring function design, and multiview contrastive learning. For fine-grained representation enhancement, we introduce a feature disentanglement module and a key term highlighting mechanism for images and text, respectively. Finally, we conduct end-to-end model training by jointly optimizing the IFDM loss and contrastive learning losses.

$H \times W$, $T^{(m)}$ stands for its corresponding textural description, and $\boldsymbol{y}^{(m)} \in \mathbb{R}^C$ denotes the one-hot vector of image labels, with C being the number of label categories. We first extract initial global representations and fine-grained representations for both images and text. Specifically, the input image is divided into a series of image patches with positional information, which are then fed into a visual encoder to generate fixed vector embeddings. The text sentence is processed through tokenization and sequentially fed into a text encoder. This process can be formulated as

$$\begin{aligned} V &= \text{encode}_v(I) \\ E &= \text{encode}_t(T) \end{aligned} \tag{1}$$

where $V = \{\boldsymbol{v_0}, \boldsymbol{v_1}, \ldots, \boldsymbol{v}_{N_v}\}$ is a feature set consisting of a global feature $\boldsymbol{v_0}$ and local image patch features $V_l = \{\boldsymbol{v_i}\}_{i=1}^{N_v} \in \mathbb{R}^{N_v \times D_v}$. $E = \{\boldsymbol{e_0}, \boldsymbol{e_1}, \ldots, \boldsymbol{e}_{N_t}\}$ represents a feature set composed of a sentence feature $\boldsymbol{e_0}$ and word features $E_l = \{\boldsymbol{e_i}\}_{i=1}^{N_t} \in \mathbb{R}^{N_t \times D_t}$ of the description. The aforementioned operations employ pretrained modal encoders encode_v and encode_t, both derived from state-of-the-art methods. A typical example is CLIP [10], which provides well-performing pretrained visual and text encoders for direct use.

C. Image Feature Disentanglement and Enhancement

To highlight essential objects in images, we first employ the squeeze-and-excitation (SE) network [45] to enhance patch representations. Then, a feature decoupling network is designed to separate object features from backgrounds. This operation effectively mitigates interference caused by large-scale background regions and noise, focusing the model on meaningful target cues. Finally, a GNN-based module strengthens the intrinsic relationships between objects, providing refined fine-grained image representations. Detailed implementation of each component is given as follows.

1) SE Network Module: We utilize the SE network to select and enhance relevant image patches, as it models interchannel correlations and adaptively captures the importance of each channel. Within this module, a multilayer perceptron (MLP) calculates and reweights features across different channels, thereby improving the expressiveness and discriminability of image features.

Specifically, the SE module enhances the image representation $V_l = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \ldots, \boldsymbol{v}_{N_v}\} \in \mathbb{R}^{N_v \times d}$, with the output denoted as $V_l' = SE(V_l)$. The complete calculation process is

$$\begin{aligned} \bar{\boldsymbol{w}} &= \text{GPOOL}(V_l) \\ \boldsymbol{w} &= \text{MLP}(\bar{\boldsymbol{w}}) \\ \boldsymbol{v_i'} &= w_i * \boldsymbol{v_i} \quad i = 1, 2, \ldots, N_v. \end{aligned} \tag{2}$$

Here, $\bar{\boldsymbol{w}} \in \mathbb{R}^{N_v}$ and $\boldsymbol{w} = (w_1, w_2, \ldots, w_{N_v}) \in \mathbb{R}^{N_v}$ are channel-wise weights, with GPOOL denoting the global pooling operation. This yields the enhanced representation $V_l' = \{\boldsymbol{v_1'}, \boldsymbol{v_2'}, \ldots, \boldsymbol{v}_{N_v}'\}$.

2) Image Feature Disentanglement Module: We propose a novel IFDM to separate background information and noises from each image patch. Specifically, we construct an autoencoder framework to learn the latent space, and jointly optimize the model with three learning tasks, that is, feature reconstruction, scene discrimination, and background information alignment, to achieve fine-grained disentanglement and image

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feature refinement. The detailed implementation is outlined below.

1) *Decoupled Representation of Features:* We adopt a visual encoder to project the global image representation $\boldsymbol{v_0}$ together with the enhanced fine-grained representations $V_l' = \{\boldsymbol{v_1'}, \boldsymbol{v_2'}, \ldots, \boldsymbol{v_{N_v}'}\}$ into a latent feature space:

$$\widetilde{V} = \text{encode}_{ifdm}\left(\{\boldsymbol{v_0}, V_l'\}\right) \tag{3}$$

where $\widetilde{V} = \{\tilde{\boldsymbol{v}}_0, \tilde{\boldsymbol{v}}_1, \tilde{\boldsymbol{v}}_2, \ldots, \tilde{\boldsymbol{v}}_{N_v}\}$, and each feature vector contains both background and object features:

$$\widetilde{V} = \left[\widetilde{V}_{bg} : \widetilde{V}_{ob}\right] \tag{4}$$

where $[:]$ denotes the concatenation operation. In this structure, the first k dimensions of each feature vector represent background information, while the remaining part corresponds to the object representation.

2) *Feature Reconstruction Task:* We reconstruct the input through a decoder and use mean squared error (mse) to measure the reconstruction losses as follows:

$$\hat{V}_{rec} = \text{decode}_{ifdm}\left(\widetilde{V}\right) \tag{5}$$

$$L_{rec_g} = \|\hat{\boldsymbol{v}}_0 - \boldsymbol{v_0}\|^2 \tag{6}$$

$$L_{rec_l} = \frac{1}{N_v - 1}\sum_{i=1}^{N_v}\left\|\hat{\boldsymbol{v}}_i - \boldsymbol{v_i'}\right\|^2 \tag{7}$$

where L_{rec_g} and L_{rec_l} denote the reconstruction losses of the global representation and local representations, respectively. In our implementation, both encode_{ifdm} and decode_{ifdm} adopt an MLP structure.

3) *Scene Label Classification Task:* Intuitively, an image's background information should reflect its scene category. We therefore introduce a label discrimination task to guide the learning of background features. For a given image, background representations of its patches, that is, $\widetilde{V}_{bg} = \{\boldsymbol{v}_{\tilde{bg}_0}, \boldsymbol{v}_{\tilde{bg}_1}, \ldots, \boldsymbol{v}_{\tilde{bg}_N_v}\}$, are used as input to a linear classifier, which predicts the corresponding scene labels $\hat{\boldsymbol{y}}_i$ (typically, each image maps to multiple scene labels, with C denoting the total number of label categories)

$$\hat{\boldsymbol{y}}_i = \text{classify}\left(\boldsymbol{v}_{\tilde{bg}_i}\right) \quad i = 0, 1, \ldots, N_v. \tag{8}$$

Subsequently, we calculate the cross-entropy losses for global and local representations, denoted as L_{cla_g} and L_{cla_l}, respectively,

$$L_{cla_g} = -\sum_{c=1}^{C} y_{0_c}\log\left(\hat{y}_{0_c}\right) \tag{9}$$

$$L_{cla_l} = -\sum_{i=1}^{N_v}\sum_{c=1}^{C} y_{i_c}\log\left(\hat{y}_{i_c}\right). \tag{10}$$

4) *Background Information Alignment Task:* Ideally, the local background information $\boldsymbol{v}_{\tilde{bg}_l}(l = \{1, 2, \ldots, N_v\})$ corresponding to different image patches should be consistent with the global background information $\boldsymbol{v}_{\tilde{bg}_0}$. Therefore, we guide feature disentanglement by minimizing the difference L_{bga} between them

$$L_{bga} = \frac{1}{N_v - 1}\sum_{i=1}^{N_v}\left\|\boldsymbol{v}_{\hat{bg}_i} - \boldsymbol{v}_{\hat{bg}_0}\right\|^2. \tag{11}$$

To ensure the effectiveness of the image disentanglement module, the losses corresponding to the three aforementioned learning tasks are jointly optimized during the model training process. Detailed content will be introduced in Section III-G.

3) GNN-Based Enhancement Module: Aiming to capture the internal relationships among objects, we construct a fully connected graph with self-loops and use a graph convolutional network (GCN) for representation learning. Following [46], we take the enhanced object features $\widetilde{V}_{ob} = \{\widetilde{\boldsymbol{v}_{ob_1}}, \widetilde{\boldsymbol{v}_{ob_2}}, \ldots, \widetilde{\boldsymbol{v}_{ob_N_v}}\}$ as the initial graph node representation, and use the cosine similarity between nodes as the edge information W. The graph updating process is defined as

$$V^* = W_{gr}\left(W\widetilde{V}_{ob}W_g\right) + \widetilde{V}_{ob} \tag{12}$$

where $W_g \in \mathbb{R}^{d\times d}$ denotes the inference weights between graph nodes, and $W_{gr} \in \mathbb{R}^{N_v\times N_v}$ is the weight matrix of the residual structure. Here, the GCN allows multilayer stacking. Since each image patch contributes differently to semantic information, we employ a gated recurrent unit (GRU) (see [47]) to generate the final hidden state as a comprehensive fine-grained embedding v_h of the entire sequence, owing to its ability to model sequential dependencies and its simple implementation. Given the input sequence V^*, v_h is calculated as

$$v_h = \text{GRU}\left(V^*\right). \tag{13}$$

Note that the number of graph nodes N_v is a constant for a given CLIP backbone, ranging from 49 to 576. Since N_v is much smaller than the dataset size, this marginal computational cost introduced by graph construction and processing can be negligible.

D. Key Semantic Enhancement for Textual Descriptions

Since descriptions associated with images are often general, key terms account for a relatively small proportion of the overall content, leaving critical details (e.g., object attributes) prone to being overshadowed. To focus on key information, we first identify semantically meaningful tokens from the text, apply feature enhancement to these tokens, and then amplify their influence in image–text matching to boost retrieval performance. In this process, we leverage the Natural Language Toolkit (NLTK) library to annotate the part-of-speech (POS) of each word, highlighting high-value words such as nouns, adjectives, and adverbs.

Given the text E_l as input, after selecting $N_{t'}$ key words, the resulting output is $E' = \{\boldsymbol{e_1}, \boldsymbol{e_2}, \ldots, \boldsymbol{e_{N_{t'}}}\}$. For instance, $E_l = \{\text{the, airport, covers, a, large, area, and, many, planes}\}$ is extracted as $E' = \{\text{airport, covers, large, area, many, planes}\}$. Next, similar to the image processing workflow, E' is fed into the SE and GRU modules to obtain the enhanced fine-grained text representation E^*

$$\hat{E} = \text{SE}\left(E'\right) \tag{14}$$

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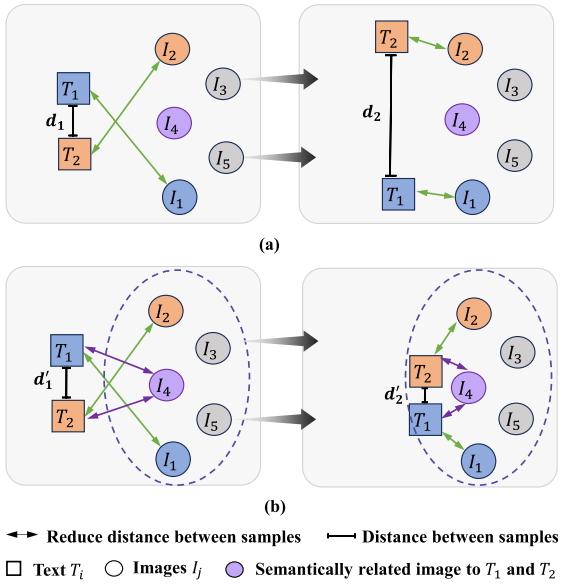


Fig. 2. Motivation of our group-aware image–text alignment mechanism. Given that T_1 and T_2 share the same label, (T_1, I_1) and (T_2, I_2) are matched pairs, while I_4 is semantically relevant to both T_1 and T_2. Compared with the instance-level alignment mechanism, our group-aware method can regulate representation distribution from a macro perspective. (a) Instance-level image–text alignment. (b) Group-aware image–text alignment.

$$E^* = \mathrm{GRU}\left(\hat{E}\right) \tag{15}$$

where $\hat{E} = \{\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \ldots, \hat{\boldsymbol{e}}_{N_{t'}}\} \in \mathbb{R}^{N_{t'} \times D}$, and $E^* = \{\boldsymbol{e}_1^*, \boldsymbol{e}_2^*, \ldots, \boldsymbol{e}_{N_{t'}}^*\} \in \mathbb{R}^{N_{t'} \times D}$.

E. Group-Aware Image–Text Similarity Function Design

Existing fine-grained matching methods primarily focus on instance-level image–text alignment, neglecting the optimization of shared representation distribution by considering global modal correlations, restricting the model to distinguish subtle intersample differences. As shown in Fig. 2(a), T_1 and T_2 share the same label, while (T_1, I_1) and (T_2, I_2) are matched pairs. The instance-level image–text alignment mechanism aims to reduce the distance between each positive pair, that is, pushing T_1 toward I_1 and T_2 toward I_2. However, without accounting for global sample relationships, the scenario illustrated in the figure may arise: the distance between T_1 and T_2 becomes inadvertently enlarged.

To address this issue, we propose a group-aware image–text similarity scoring function, designed to optimize the shared representation space from a macro perspective with two core motivations. First, we argue that each sample should not only move closer to its matched counterpart but also align with semantically relevant group. As illustrated in Fig. 2(b), since I_4 is semantically relevant to both T_1 and T_2, we regard I_4 as an essential anchor to guide the optimization of cross-modal representations, ensuring comprehensive capture of both intramodal and intermodal relationships. Second, to enhance the model's ability to distinguish subtle semantic gaps between samples, we propose to identify anchors based on the distinct key semantics of the query, and then compute the group-aware similarity score s_h using the distance between the query's semantic set and the reference group.

To show the calculation of s_h, let us take the text-to-image retrieval task as an example. As presented in Fig. 1, we use the image set $B_v = \{\boldsymbol{v}_{h_1}, \boldsymbol{v}_{h_2}, \ldots, \boldsymbol{v}_{h_B}\}$ in the current training batch as a shared reference group. Meanwhile, $N_{t'}$ key words are selected from the given text, with their enhanced fine-grained representations denoted as $E^* = \{\boldsymbol{e}_1^*, \boldsymbol{e}_2^*, \ldots, \boldsymbol{e}_{N_{t'}}^*\}$. Next, we decompose the text semantics and select a set of image instances from dataset B_v with the strongest semantic relevance to the decomposed semantics of E^*. The average cosine similarity of these instances is then calculated as the group-aware similarity score s_h, denoted as

$$s_h = \frac{1}{N_{t'}} \sum_{i=1,\ldots,N_{t'}} \max_{j=1,\ldots,B} \cos\left(\boldsymbol{e}_i^*, \boldsymbol{v}_{\mathbf{h}_j}\right). \tag{16}$$

In addition to s_h, we also calculate the global semantic similarity score s_g between the image and text to ensure the basic modal matching capability. Furthermore, since our method freezes the CLIP backbone network to enable lightweight model training, we introduce a probe similarity score s_p via an MLP to expand the model's learning and representation spaces. Formally, the global semantic similarity score s_g is given by

$$s_g = \cos\left(\boldsymbol{e}_{\mathbf{0}}, \boldsymbol{v}_{\mathbf{0}}\right) \tag{17}$$

where $\boldsymbol{e}_{\mathbf{0}}$ and $\boldsymbol{v}_{\mathbf{0}}$ represent the global representations (or coarse-grained semantics) of text and image, respectively, while cos refers to the cosine similarity calculation.

The probe similarity score s_p for this component is calculated as

$$\dot{\boldsymbol{e}}_0 = W_e \boldsymbol{e}_{\mathbf{0}} + \boldsymbol{b}_e \tag{18}$$

$$\dot{\boldsymbol{v}}_0 = W_v \boldsymbol{v}_{\mathbf{0}} + \boldsymbol{b}_v \tag{19}$$

$$s_p = \cos\left(\dot{\boldsymbol{e}}_0, \dot{\boldsymbol{v}}_0\right). \tag{20}$$

Here, W and b denote the weight and bias parameters of the linear mapping, respectively.

By synthesizing the three aspects mentioned above, the final modal similarity scoring function is given by

$$s = \lambda_g s_g + \lambda_p s_p + \lambda_h s_h \tag{21}$$

where λ_g, λ_p and λ_h are hyperparameters satisfying $\lambda_g + \lambda_p + \lambda_h = 1$.

Our scoring function enables key text components to accurately match corresponding image objects, while narrowing the distance between text and similar images during training. It regularizes the shared representation space from a macro perspective and strengthens discriminative power: using images as References decomposes the text's fine-grained semantics, amplifies textual differences, and assesses image–text pair relationships from multiple views, thus encouraging our model to distinguish subtle semantic variations among samples.

F. Multiview Contrastive Learning

Contrastive learning pulls positive pairs closer and pushes negative pairs apart, yielding a high-quality feature distribution. In our scheme, we fully consider the correlation between

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individuals and groups, and design a multiview contrastive learning strategy to optimize the shared representation space. In addition to instance-level contrastive learning, we use "semantically related samples" to the query as a reference: for similar samples, their distances to the reference group should be close; if they form a negative pair, the difference in their distances to the group should be more pronounced.

Formally, we use the triplet ranking loss to directly optimize the distribution of shared representation space from the group perspective, constraining the similarity of positive pairs to exceed that of negative pairs. Given a training batch $(\mathcal{I}, \mathcal{T})$ containing N image–text pairs, the loss function L_{cl_g} is expressed as

$$L_{cl_g} = \sum_{I\in\mathcal{I}} \max\left(\varepsilon - s\left(I,T\right) + s\left(I,T^{-}\right),0\right) + \sum_{T\in\mathcal{T}} \max\left(\varepsilon - s\left(I,T\right) + s\left(I^{-},T\right),0\right) \tag{22}$$

where ε is a margin value, T^{-} and I^{-} denotes the hard negative samples corresponding to I and T in the training batch, respectively. Since the group-aware similarity score $s_h(\cdot)$ is a component of $s(\cdot)$, the above contrastive learning strategy can encourage each text to align with semantically related image groups, and vice versa.

Furthermore, for the image–text data pairs within the batch, we adopt the InfoNCE loss L_{cl_s} for single-instance contrastive learning based on the extended probe similarity score s_p from the individual perspective

$$L_{cl_s} = \frac{1}{2}\left(L_{i2t} + L_{t2i}\right) \tag{23}$$

where the image-to-text (I2T) contrastive loss L_{i2t} and the text-to-image contrastive loss L_{t2i} are calculated by

$$L_{i2t} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(s_p\left(I^{(i)},T^{(i)}\right)/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_p\left(I^{(i)},T^{(j)}\right)/\tau\right)} \tag{24}$$

$$L_{t2i} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(s_p\left(I^{(i)},T^{(i)}\right)/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_p\left(I^{(j)},T^{(i)}\right)/\tau\right)}. \tag{25}$$

G. Loss Function and Model Training

We adopt an end-to-end model training paradigm to jointly optimize the image feature disentanglement loss L_{dis} and the multiview contrastive learning loss defined in Section III-F. The overall loss function L is formulated as

$$L = \alpha L_{cl_g} + \beta L_{cl_s} + \gamma L_{dis} \tag{26}$$

where the composition of each loss term is

$$L_{dis} = L_{rec} + L_{cla} + L_{bga} \tag{27}$$

$$L_{rec} = \gamma_g L_{rec_g} + \gamma_l L_{rec_l} \tag{28}$$

$$L_{cla} = \gamma_g L_{cla_g} + \gamma_l L_{cla_l}. \tag{29}$$

Here, $\alpha, \beta, \gamma, \gamma_g$, and γ_l are all hyperparameters. Considering the difference in magnitude between global and local losses during the image disentanglement process, we assign distinct weights γ_g and γ_l (with $\gamma_g + \gamma_l = 1$) to ensure the dominant position of global features.

IV. Experiments

A. Experimental Settings

1) Datasets: Extensive experiments have been conducted on two publicly accessible remote sensing datasets: the remote sensing image captioning dataset (RSICD) [49] and the remote sensing image–text match dataset (RSITMD) [12]. RSICD consists of 10 921 images (224 × 224 pixels), each with five descriptions. These images were collected from Google Earth, Baidu Map, MapABC, and Tianditu, covering 30 categories. RSITMD comprises 4743 images (256 × 256 pixels), each also with five descriptive sentences. It covers 32 categories, including two new categories (*boat* and *intersection*) compared to RSICD. In our experiments, consistent with the approach in [12], both datasets were partitioned into training, validation, and testing sets in an 8:1:1 ratio.

2) Evaluation Metrics: We employ two widely recognized metrics to evaluate the model's performance: recall at k (R@k) for $k \in \{1, 5, 10\}$ and mean recall (mR). Specifically, R@k denotes the proportion of positive results that appear within the top-k retrieved items for a given query, while mR represents the average of R@k values across different k settings. Formally, these two metrics are defined as follows:

$$\text{R@}k = \frac{1}{N}\sum_{i=1}^{N} I\left(i,k\right) \tag{30}$$

$$\text{mR} = \frac{1}{6}\sum_{k\in\{1,5,10\}} \left([\text{R@}k]_{i2t} + [\text{R@}k]_{t2i}\right). \tag{31}$$

Here, N represents the total number of queries, and $I(i,k)$ is an indicator function that equals 1 if the relevant result for the i-th query is among the top-k retrieved items, and 0 otherwise. Additionally, *i2t* refers to I2T retrieval, while *t2i* denotes text-to-image retrieval, where a higher recall value indicates superior performance.

3) Comparison Algorithms: To validate the effectiveness of our proposed method, we compare it against a range of state-of-the-art cross-modal retrieval methods tailored for remote sensing scenarios. These methods encompass both deep learning-based approaches (e.g., AMFMN [12], GaLR [13], KAMCL [21], DOVE [23], MSITA [22], and PIR [24]), and VLP-based methods (e.g., IEFT [34], PETL [40], RemoteCLIP [42], CLGSA [36], FGVLA [37], GLISA [43], and CLIP [10]). All models were implemented using PyTorch with Python 3.8 and trained on a single NVIDIA RTX A6000 48GB GPU. To ensure a fair comparison, we maintained consistent values for shared parameters across all methods while setting the remaining parameters in accordance with the optimal values reported in their respective original papers.

4) Implementation Details: For our proposed method, the image–text encoder is built on CLIP (ViT-B/16). The hyperparameters for feature decoupling, namely γ_g and γ_l, are set to 0.95 and 0.05, respectively. On the RSICD dataset, the weights within the score function, namely λ_g, λ_p, and λ_h, are configured as 0.2, 0.3, and 0.5, respectively. The parameters in the final loss function, that is, α, β, and γ, are set at a ratio of 2:1:1. Similarly, on the RSITMD dataset, the score function weights λ_g, λ_p, and λ_h are set to 0.3, 0.2, and 0.5, respectively.

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TABLE I

ALGORITHM COMPARISON ON THE RSICD DATASET. CLIP FT REFERS TO THE FINE-TUNED VERSION OF CLIP. THE BACKBONE OF OUR METHOD IS FROZEN, AND ONLY A SMALL NUMBER OF NETWORK PARAMETERS NEEDS TO BE UPDATED DURING MODEL TRAINING

Methods	Reference	Backbone (image)	Params (M)	Image-to-Text retrieval			Text-to-Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional deep learning-based methods										
AMFMN	*TGRS'22*	Resnet18	35.94	5.21	14.72	21.57	4.08	17.00	30.60	15.53
GaLR	*TGRS'22*	Resnet18	46.89	6.59	19.85	31.04	4.69	19.48	32.13	18.96
KAMCL	*TGRS'23*	Resnet50	40	12.08	27.26	38.70	8.65	27.43	42.51	26.10
DOVE	*TGRS'24*	Resnet50	38	8.66	22.35	34.95	6.04	23.95	40.35	22.71
MSITA	*TGRS'24*	Resnet+Transformer	48.14	8.67	22.71	33.91	6.13	21.98	35.39	21.47
PIR	*ACM MM'23*	Swin-Transformer	140.5	9.88	27.26	39.16	6.97	24.56	38.92	24.46
VLP-based methods										
IEFT	*TGRS'23*	ViLT	100.12	8.38	28.17	44.16	7.78	28.47	43.88	26.81
PETL	*TGRS'23*	ViT-B/32	0.16	14.13	31.51	44.78	11.63	33.92	50.73	31.12
RemoteClip	*TGRS'24*	ViT-B/32	87	17.02	37.97	51.51	13.71	37.11	54.25	35.26
CLGSA	*TGRS'25*	ViT-B/16	125.6	14.38	36.73	45.35	11.75	34.07	50.12	32.07
FGVLA	*TGRS'25*	ViT-B/16	138.2	16.93	36.23	48.49	12.72	36.61	53.76	34.12
GLISA	*TGRS'24*	ViT-B/16	154.8	19.52	40.44	52.28	14.75	39.50	55.46	36.99
GLISA+ours	-	ViT-B/16	**4.51**	**20.80**	**42.60**	**54.70**	**15.80**	**41.40**	**58.30**	**38.93**
CLIP*ft*	*ICML'21*	ViT-B/16	151	18.80	39.50	54.00	15.40	41.10	57.90	37.78
CLIP*ft*+ours	-	ViT-B/16	**4.51**	**22.00**	**44.60**	**55.40**	**15.60**	**41.80**	**58.80**	**39.70**

TABLE II

ALGORITHM COMPARISON ON THE RSITMD DATASET. CLIP FT REFERS TO THE FINE-TUNED VERSION OF CLIP. THE BACKBONE OF OUR METHOD IS FROZEN, AND ONLY A SMALL NUMBER OF NETWORK PARAMETERS NEEDS TO BE UPDATED DURING MODEL TRAINING

Methods	Reference	Backbone (image)	Params (M)	Image-to-Text retrieval			Text-to-Image retrieval			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
Traditional deep learning-based methods										
AMFMN	*TGRS'22*	Resnet18	35.94	10.63	24.78	41.81	11.51	34.69	54.87	29.72
GaLR	*TGRS'22*	Resnet18	46.89	14.82	31.64	42.48	11.15	36.68	51.68	31.41
KAMCL	*TGRS'23*	Resnet50	40	16.51	36.28	49.12	13.50	42.15	59.32	36.14
DOVE	*TGRS'24*	Resnet50	38	16.81	36.80	49.93	12.20	44.13	66.50	37.73
MSITA	*TGRS'24*	Resnet+Transformer	48.14	15.22	34.20	47.65	12.15	39.93	57.72	34.48
PIR	*ACM MM'23*	Swin-Transformer	140.5	18.14	41.15	52.88	12.17	41.68	63.41	38.24
VLP-based methods										
IEFT	*TGRS'23*	ViLT	100.12	11.19	38.09	58.84	15.49	37.61	51.40	35.44
PETL	*TGRS'23*	ViT-B/32	0.16	23.67	44.07	60.36	20.10	50.63	67.97	44.47
RemoteClip	*TGRS'24*	ViT-B/32	87	27.88	50.66	65.21	22.17	56.16	73.41	49.24
CLGSA	*TGRS'25*	ViT-B/16	125.6	26.77	48.23	60.39	22.52	52.39	69.46	46.63
FGVLA	*TGRS'25*	ViT-B/16	138.2	26.33	51.99	64.38	21.77	54.12	72.92	48.59
GLISA	*TGRS'24*	ViT-B/16	154.8	28.21	50.94	62.50	20.50	54.80	72.19	48.19
GLISA+ours	-	ViT-B/16	**4.51**	**29.00**	**53.50**	**63.90**	**22.90**	**56.70**	**73.90**	**49.98**
CLIP*ft*	*ICML'21*	ViT-B/16	151	28.10	51.50	62.40	22.60	56.20	73.60	49.07
CLIP*ft*+ours	-	ViT-B/16	**4.51**	**31.00**	**53.80**	**65.53**	**23.10**	**56.40**	**74.10**	**50.66**

The loss function ratios α, β, and γ are set to 1.0:1.5:1.0. The training batch size is 256, and the Adam optimizer is used with a learning rate of 5×10^{-4}.

B. Overall Performance

Note that our method can be flexibly integrated into the existing dual-tower image–text alignment paradigm. With the backbone frozen, only a few parameters require training. Tables I and II summarize the performance of all compared methods on the two remote sensing datasets. From these experimental results, we can derive the following key conclusions.

1) Our proposed method demonstrates consistent and significant improvements when integrated into two strong baselines, that is, GLISA and CLIP*ft*, achieving new state-of-the-art performance on the RSICD and RSITMD datasets. Specifically, on RSICD, our method brings a 1.94% mR improvement to GLISA and a 1.92% gain to CLIP*ft*, attaining the highest mR of 39.70. On RSITMD, it elevates GLISA by 1.79% and CLIP*ft* by 1.59% in mR, reaching 50.66%. Notably, for R@1 in I2T retrieval, our approach boosts GLISA by 1.28% and CLIP*ft* by 3.20% on RSICD, and by 0.79% and 2.90%, respectively, on RSITMD. This indicates that by enhancing fine-grained feature representation to capture discriminative semantics and strengthening multiperspective matching, our method improves the cross-modal discriminative ability, thereby effectively boosting retrieval performance.
2) We observe consistent performance disparities across both datasets: compared to CLIP*ft*, our method demonstrates greater improvements in I2T retrieval (with gains

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of 9.7% and 8.3%) than in text-to-image retrieval (with gains of 1.8% and 1.2%). This asymmetry stems from the semantic enhancement mechanism embedded in visual representations, which more effectively aligns discriminative image regions with textual descriptions. Conversely, text queries inherently contain structured semantic information, leaving less optimization headroom in complex visual scenes.

3) Given that our model is a lightweight architecture built on a frozen LLM (e.g., CLIP*ft*), it entails significantly lower parameter counts and training costs. For instance, it requires only 3.0% of the parameters used by CLIP*ft* and 2.9% of those by GLISA. In terms of training time, fine-tuning CLIP on the RSICD dataset takes 3.67 hours, whereas our method requires merely 0.83 hours, demonstrating a distinct advantage in training efficiency.
4) Fig. 3 presents the retrieval results of CNN-based and transformer-based methods. It reveals a fundamental advantage of transformer-based backbones over CNN architectures. This superiority stems from the global attention mechanisms of transformers, which model cross-modal dependencies holistically. In contrast, CNNs exhibit inductive bias toward local feature extraction which limits semantic alignment in vision-language tasks.
5) Fig. 3 compares VLP-based methods with deep learning-based approaches, revealing the former's comprehensive superiority. This originates from large-scale pretrained models, which inherently encode cross-modal alignment knowledge via large-scale multimodal training. These models enable efficient adaptation to new domains via zero-shot transfer learning, eliminating the need for laborious hand-crafted feature extraction pipelines.

C. Ablation Study

1) Ablation Study on Components of FGR-GA: We have conducted ablation experiments to evaluate the effects of the IFDM, the key word selection (KWS) operation, and group-aware contrastive learning (GACL) module. Specifically, we construct multiple variants of the proposed FGR-GA method and assess their query accuracy. All variants adopt CLIP (ViT-B/16 version) as the backbone, with mean recall rates of text-to-image and I2T retrieval under $k \in \{1, 5\}$ used as evaluation metrics.

1) *Without GACL* denotes the setting where the group-aware similarity score s_h is removed from the final modal similarity scoring function [see (21)] and the contrastive learning loss L_{gcl} [see (22)], respectively.
2) *Without IFDM* refers to the setting where the IFDM component is replaced with a simple MLP module.
3) *Without KWS* stands for the setting where no keyword selection is applied to the text, that is, using E instead of E' in (14).
4) *Base* represents the baseline configuration with both the IFDM and GACL modules removed from our model.
5) *FGR-GA* corresponds to our complete model.

As shown in Table III, the KWS, IFDM, and GACL modules significantly enhances cross-modal retrieval performance.

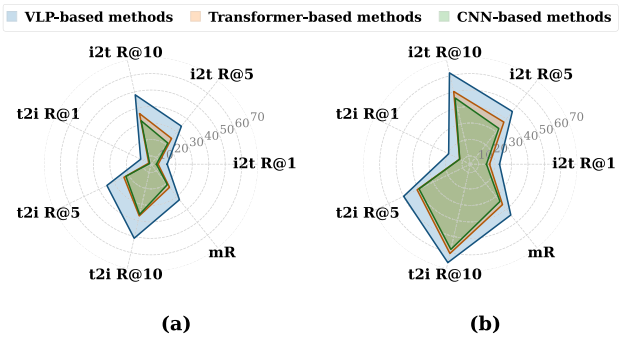


Fig. 3. Performance comparison of various remote sensing image–text retrieval methods. Average retrieval accuracy corresponding to each type of method are calculated from Tables I and II. (a) RSICD. (b) RSITMD.

TABLE III
ABLATION RESULTS ON THE RSICD AND RSITMD DATASETS

Methods	RSICD			RSITMD		
	mR@1	mR@5	mR	mR@1	mR@5	mR
Base	17.20	40.30	37.78	25.35	53.85	49.07
w/o GACL	17.45	40.80	38.10	26.05	54.20	49.77
w/o IFDM	18.70	42.50	39.35	26.35	54.45	50.18
w/o KWS	18.77	42.85	39.58	26.90	55.10	50.52
FGR-GA (ours)	18.80	43.20	39.70	27.05	55.10	50.66

TABLE IV
ABLATION STUDY ON COMPONENTS OF L_{dis}

Methods			RSICD			RSITMD		
L_{rec}	L_{bga}	L_{cla}	mR@1	mR@5	mR	mR@1	mR@5	mR
-	-	-	18.70	42.50	39.35	26.35	54.45	50.18
✓	-	-	18.51	43.04	39.53	26.33	54.89	50.13
-	✓	-	19.01	42.77	39.49	27.04	53.83	50.14
-	-	✓	18.77	43.16	39.53	26.77	54.76	50.30
✓	✓	-	18.79	43.07	39.58	27.35	53.81	50.09
✓	-	✓	18.66	42.89	39.43	27.06	54.25	50.30
-	✓	✓	18.77	43.21	39.69	26.84	54.47	50.22
✓	✓	✓	18.80	43.20	39.70	27.05	55.10	50.66

Compared with FGR-GA, the comprehensive performance gains achieved by the GACL-enabled variant (i.e., w/o GACL) demonstrate how contrastive learning optimizes matching relationships from a macro perspective. The variant incorporating IFDM (i.e., w/o IFDM) achieves improvements in both mR@1 and mR@5 metrics, validating the efficacy of the IFDM in capturing fine-grained visual patterns and precise cross-modal alignment. While IFDM and KWS only bring moderate gains when deployed individually, they can effectively cooperate with GACL to significantly improve cross-modal matching quality. Specifically, the disentangled object features generated by IFDM serve as geometrically stable anchors for GACL's contrastive optimization, while keyword selection enhances the model's robustness to ambiguous and noisy descriptions, collectively enhancing the discriminative capability of our model.

2) Ablation Study on Components of L_{Dis}: As shown in Table IV, we conduct ablation experiments on components of L_{dis}, that is, L_{rec}, L_{bga}, and L_{cla} [see (27)]. Experimental results demonstrate that all three losses are critical components,

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TABLE V

COMPARISON OF MODEL TRAINING AND INFERENCE COSTS ON RSITMD. THE TRAINING TIMES FOR H/14 AND G/14 ARE ESTIMATED FROM PER-BATCH DURATION MULTIPLIED BY TEN EPOCHS. THE BACKBONE OF FGR-GA IS CLIP-B/16

Methods	CLIP-B/32	CLIP-B/16	CLIP-L/14	CLIP-L/14@336px	CLIP-H/14	CLIP-G/14	FGR-GA (Ours)
Training Time (min)	9	22	86	196	345	490	5
Batchsize	256	128	32	16	8	8	256
GPU Memory Usage (M)	26318	33802	33596	44484	25026	33498	7020
Parameters (M)	144.2	142.6	407.8	408.1	940.4	1303.3	4.5
FLOPs (G)	4.88	13.21	56.26	120.87	123.49	193.57	13.21+0.43
Inference Latency (ms)	14.03	14.62	19.84	25.65	47.53	66.26	14.62+4.07

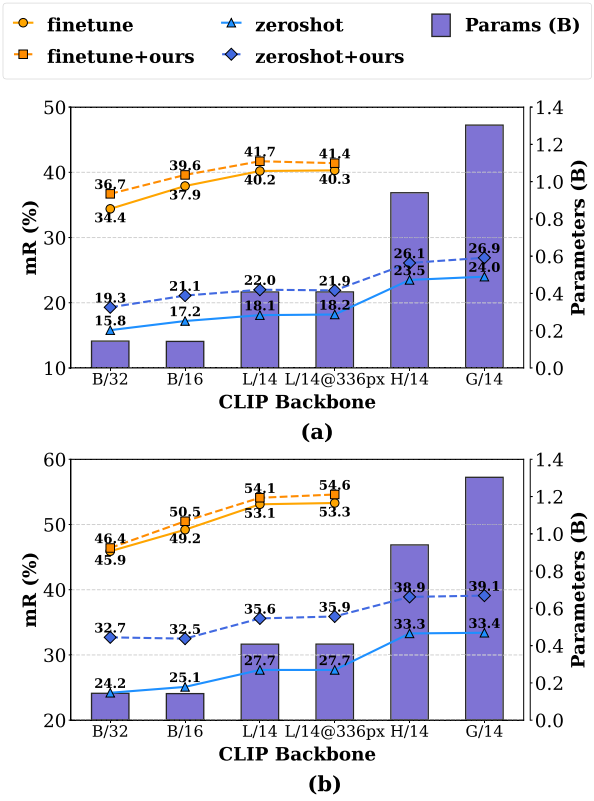


Fig. 4. Retrieval performance and parameters across different CLIP backbones. The solid line denotes results using CLIP alone, while the dashed line represents results with our proposed method incorporated. (a) Results on the RSICD dataset. (b) Results on the RSITMD dataset.

and their synergistic effect significantly boosts retrieval performance. Among scenarios with a single loss enabled (first four rows), L_{cla} delivers the most notable performance gain. However, when L_{rec} or L_{bga} is introduced alone or paired with L_{cla}, no effective performance improvement is observed. In contrast, the joint action of all three losses (last row) yields the best performance: the mR on RSITMD reaches 50.66, a 0.48 improvement over the baseline (all losses disabled), confirming the necessity of multiloss synergy for guiding image representation learning. In summary, L_{cla} forms the foundation for representation learning, while L_{rec} and L_{bga} complement it by guiding the learning of the AutoEncoder, collectively driving performance improvement.

D. Evaluation on Various Backbone Networks

To validate the generalizability of our method, we conducted comprehensive comparisons between our FGR-GA method and CLIP variants of different backbones across the RSICD and RSITMD datasets. As shown in Fig. 4, our method achieves consistent performance improvements over all baseline architectures, with relatively stable enhancement magnitudes despite increased backbone parameters. Notably, for the H/14 and G/14 backbones (with over 500M parameters), only zero-shot results are reported due to impractical memory constraints during fine-tuning (insufficient batch sizes and marginal performance gains). This analysis reveals diminishing returns of CLIP in remote sensing visual-language retrieval: performance improvements gradually plateau with increasing backbone scales. This phenomenon indicates that indiscriminate parameter expansion fails to address CLIP's inherent limitation in capturing fine-grained semantic relationships within domain-specific data, which aligns with the hypothesis proposed in the introduction.

E. Computational Cost Analysis

As shown in Table V, we evaluated the computational overhead of competing methods on the RSITMD dataset, including training time, GPU memory, floating point operations (FLOPs), and inference latency. All experiments were conducted on a single RTX A6000 (48G) GPU with optimal batch sizes. As our method freezes the CLIP backbone during training, only a small number of model parameters are updated, resulting in remarkable efficiency advantages. For instance, our CLIP-B/16-based FGR-GA trains in just 5 min, significantly faster than other CLIP variants, which range from 9 to 490 min. Although extra network layers bring slight additional overhead in inference latency and FLOPs, this is acceptable given the substantial gain in retrieval accuracy.

F. Hyperparameter Analysis

1) Effects of Batch Size: The group-aware similarity score depends on both group size and intragroup data diversity. Fig. 5 presents retrieval performance under different batch sizes with random training sampling. RSICD and RSITMD contain 30 and 32 categories, respectively. Experimental results reveal that RSICD achieves optimal performance when batch size ranges from 64 to 256, while RSITMD attains peak performance from 12 to 256. In this range, the batch-to-category ratio (≥ 2) ensures sufficient intragroup diversity for reliable group-aware similarity comparison. When the batch size is ≤ 32, significant performance degradation occurs in both datasets, primarily due to two factors: 1) insufficient group quality undermines the validity of the group-aware score and 2) inadequate negative samples impair contrastive feature

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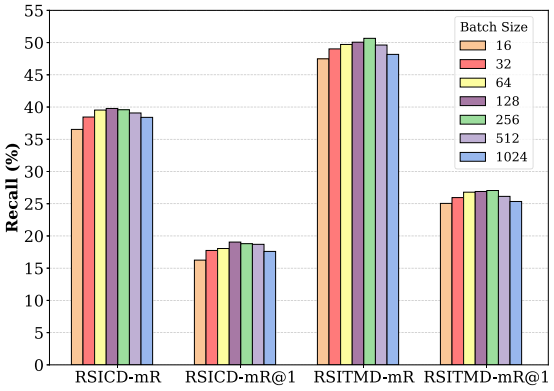


Fig. 5. Retrieval performance (mR and mR@1) under different batch sizes.

TABLE VI
RETRIEVAL PERFORMANCE (MR AND MR@1) WITH DIFFERENT DIMENSION PROPORTIONS OF BACKGROUND REPRESENTATION

Dataset-Metrics	Background Dimension Proportion					
	1/12	1/6	2/6	3/6	4/6	5/6
RSICD-mR	39.58	39.47	39.46	39.41	39.25	39.26
RSICD-mR@1	18.80	18.94	18.56	18.38	18.32	18.26
RSITMD-mR	50.66	50.32	50.42	50.41	50.30	50.19
RSITMD-mR@1	27.05	26.99	27.50	26.95	26.79	26.64

learning. Performance also declines when the batch size ≥ 512, which likely arises from excessive intragroup noise that interferes with reliable similarity evaluation. Notably, RRSITMD degrades more severely under larger batch sizes (512 and 1024), which may be attributed to its finer grained textual descriptions, leading to lower noise robustness compared with RSICD.

2) Effects of the Dimensions of Image Background Feature: We have investigated the impact of the proportion of background feature dimension to the total image feature dimension on retrieval performance, with the total feature dimension fixed at 384. On RSICD, the optimal background dimension proportion ranges from 1/12 to 2/6, while for RSITMD, this ranges from 1/12 to 3/6. As shown in Table VI, a consistent trend can be observed across both datasets: first, the model exhibits low sensitivity to the background dimension proportion, which stems from the redundancy of the total feature dimension that ensures sufficient semantic representation. However, as the background dimension proportion gradually increases, the allocation of feature resources for object representation becomes insufficient, which weakens the overall feature expression and leads to a gradual decline in retrieval performance. In our implementation, it is set to 1/12.

3) Effects of Weights in the Score Function: As shown in Fig. 6, the heatmap analysis reveals distinct optimal parameter configurations for each dataset. On RSICD, the peak performance occurs at $\lambda_g = 0.3, \lambda_p = 0.5$, and $\lambda_h = 0.2$, while RSITMD achieves its best result with $\lambda_g = 0.2, \lambda_p = 0.5$, and $\lambda_h = 0.3$. Both datasets exhibit low parameter sensitivity, particularly evident in RSITMD where mR fluctuates within 0.8% across adjacent parameter combinations

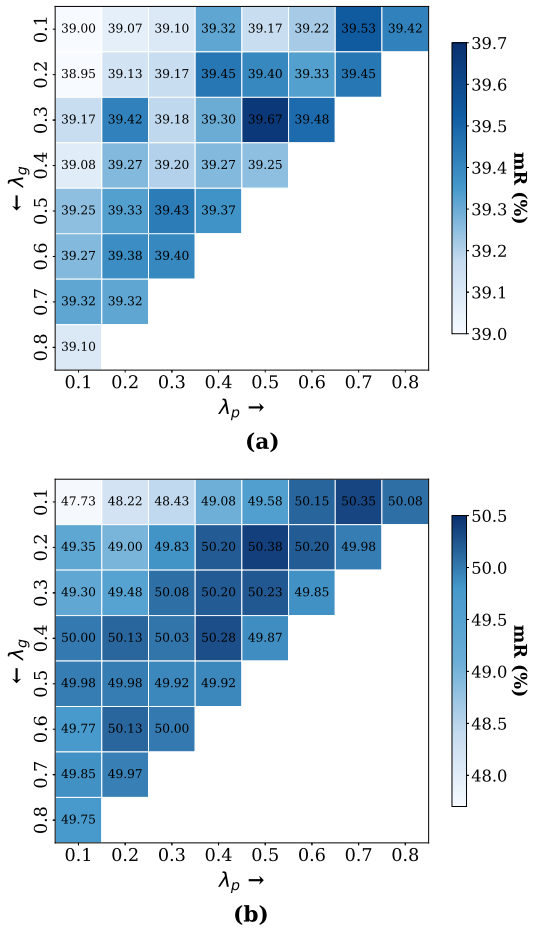


Fig. 6. Retrieval performance under different scoring parameter configurations with $\lambda_p + \lambda_g + \lambda_h = 1$ constraint. The white part represents the proportion that does not meet the constraints. (a) Results on the RSICD dataset. (b) Results on the RSITMD dataset.

$(\Delta\lambda_p \pm 0.1, \Delta\lambda_h \pm 0.1)$. Performance analysis demonstrates superior results when $\lambda_p > \lambda_h$, as the GACL module critically enhances cross-modal alignment. However, excessively small λ_h values (e.g., <0.15) degrade performance by 1.2%–1.8%, indicating that adequate dimensionality preservation in probe score s_p prevents model capacity over-constriction. Notably, λ_g variations cause minimal impact, suggesting s_g provides foundational guidance while adaptively rebalancing the ratio of λ_p and λ_h to optimize task-specific interactions.

4) Effects of Weights in the Loss Function: As shown in Fig. 7, we employ a phased parameter search strategy $\alpha \rightarrow \beta \rightarrow \gamma$ to identify optimal configurations for remote sensing cross-modal retrieval: $\alpha = 1.0, \beta = 1.0, \gamma = 1.0$(mR = 50.52%) on RSITMD and $\alpha = 2.0, \beta = 1.0, \gamma = 1.0$(mR = 39.58%). We notice that α serves as an important anchoring role, while β and γ exhibit randomness within their respective ranges, and all three parameters are highly robust.

To determine the optimal balance between the global and local alignment losses, we conduct a sensitivity analysis on the weighting coefficients γ_g (for global loss) and γ_l (for local loss), as shown in Fig. 8. Specifically, we fix one coefficient to 1 and sweep the other over an approximately logarithmic scale, which effectively covers the low-weight-sensitive region

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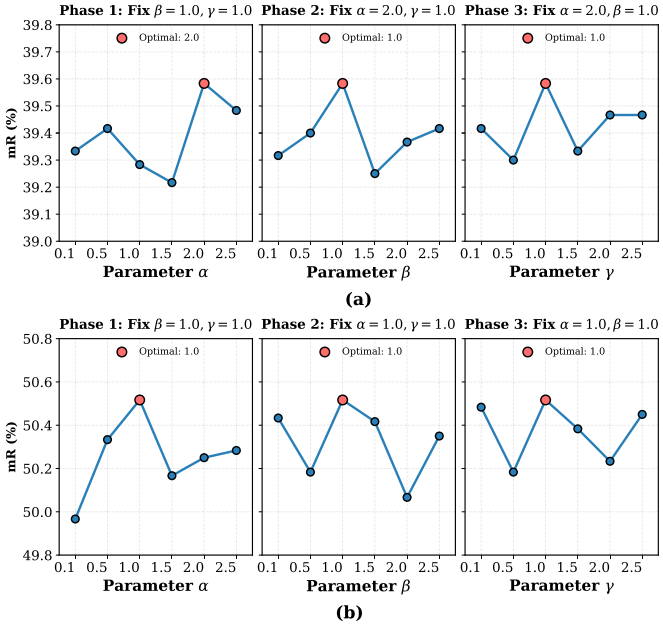


Fig. 7. Retrieval performance under the phased parameter search strategy $\alpha \rightarrow \beta \rightarrow \gamma$ in loss function on two datasets. (a) Results on the RSICD dataset. (b) Results on the RSITMD dataset.

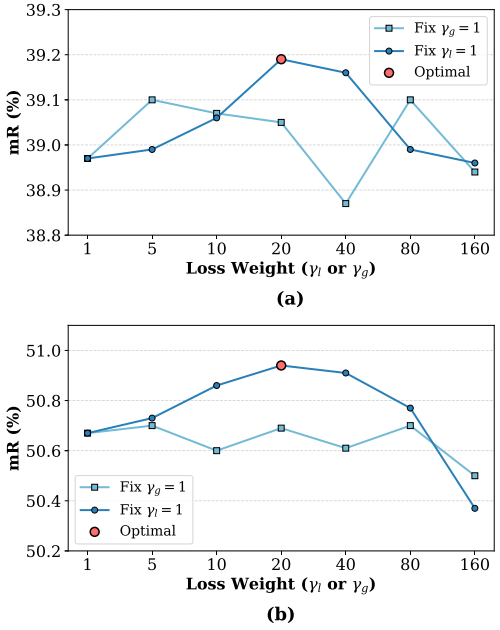


Fig. 8. Retrieval performance under the phased parameter search strategy. (a) RSICD. (b) RSITMD.

(1–20) while also revealing potential saturation effects at higher values (over 80). As shown in the figure, the model achieves peak performance when $\gamma_g = 20$ and $\gamma_l = 1$, which we adopt as the final configuration. Notably, this indicates that the global alignment loss should be assigned a relatively higher weight compared to the local counterpart, underscoring its dominant role in driving cross-modal semantic alignment for remote sensing image–text retrieval.

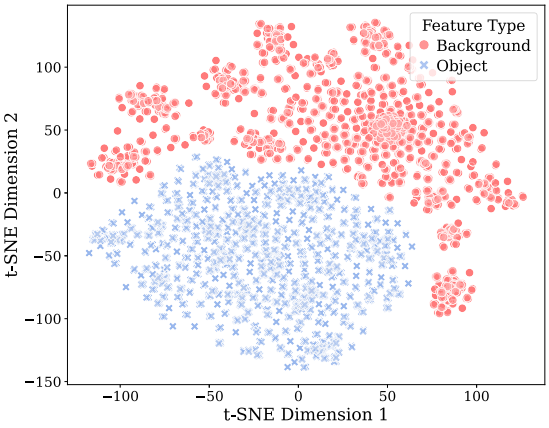


Fig. 9. Visualization of object and background features decoupled by IFDM.

G. Visualization Analysis for IFDM

We uniformly sampled a subset from the RSITMD dataset, and used t-SNE to visualize the distribution of background and object features decoupled by the IFDM module. As shown in Fig. 9, object (blue) and background (red) representations exhibit a clear distribution boundary, which directly validates the effectiveness of the decoupling module, verifies that it separates image content into semantically independent feature spaces. Meanwhile, background features show obvious clustering structures, confirming that the classification loss effectively guides background feature learning. The unclear clustering of some background features may result from the visual similarity between certain categories in remote sensing images, such as "medium residential" and "dense residential."

H. Case Study

To qualitatively validate the effectiveness of our method, we conducted case studies from the I2T perspective, analyzing several samples in Fig. 10 where our approach successfully retrieves results while CLIP*ft* fails. Comparative analysis reveals that our method achieves superior cross-modal alignment by accurately distinguishing visually similar instances and elevating matching candidates to higher rankings. Specifically, our framework demonstrates enhanced capabilities in four critical aspects: 1) precise counting of foreground objects (e.g., "six tennis courts"); 2) fine-grained recognition of small-scale targets (e.g., "red roofs and blue roofs"); 3) geometry-aware semantic detection (e.g., "chessboard" patterns and "triangle" formations); and 4) improved resistance to hallucination, effectively suppressing nonexistent semantic descriptions in query images. These representative cases confirm that our feature disentanglement mechanism and hierarchical matching paradigm significantly enhance the visual grounding accuracy of critical foregrounds and subtle targets, thereby advancing cross-modal retrieval accuracy through domain-adaptive feature representation and granularity-aware relationship modeling.

We also present failure cases in Figs. 11 and 12 to analyze typical retrieval difficulties. Notably, such challenges are shared by both our method and all baselines. The main errors arise from dataset limitations, especially coarse textual

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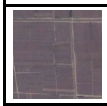
Query	Top-5 Retrieval Results on Ours	Top-5 Retrieval Results on CLIP*ft*
	There are **six tennis courts** next to the baseball field. **Six tennis courts** are next to the baseball field. A row of tennis courts are near a baseball field. There is a small tennis court surrounded by plants. There is a small tennis court with some plants.	A building with a tree is near a baseball field. The gray house met the green plants beside the red and yellow baseball field. The gray house found the green plants beside the red and yellow baseball field. The gray house met with green plants beside the red and yellow baseball field. This squared baseball field sits alongside a row of trees.
	A highway bridge connects the two sides of the river which has several houses with **red roofs and blue roofs**. There is a bridge on the river and there are buildings near it. There is a bridge on the river and there are some buildings near it. Some buildings and green plants are in two sides of a river with a bridge over it. A bridge across the river connecting two living area.	On one side of the river is a blue roof. A bridge across the river connecting two living area. The resort is located on an island with bridges. There is a row of blue roof houses on the river. The resort is located on the bridge and island.
	We can see the brown striped farmland with **netlike roads** on it. Rows of trees divide the Brown Striped farmland into squares. Dark red farmland is divided like a **chessboard**. The dark red agricultural land is divided like a **chess board**. The line of trees divided striped Brown farmland into several squares.	The line of trees divided striped Brown farmland into several squares. Rows of trees divide the Brown Striped farmland into squares. Dark red farmland is divided like a chessboard. This is a tract of bareland where there are basins of maroon soil. The crimson farmland is divided like a chessboard.
	White and grey plants are located in the **industrial triangle** between two roads. A factory with **a parked car** is located in **the triangular area** surrounded by two roads and railways. **The parking factory** is located in **a triangle** surrounded by two roads and a railway. A residential area a factory and a parking lot with tree belts are separated by a river and a road. Suburban industries include many workshops.	A parking lot with a factory area and a wooden belt is separated by the river and the road. White and grey plants are located in the industrial triangle between two roads. A residential area a factory and a parking lot with tree belts are separated by a river and a road. Road through another factory. The industrial houses on the street are in good order.
	These long, narrow ridges are covered with snow. White snow covers the peaks of the bald mountains. These long narrow ridges are covered with white snow. It flows through a river on the mountain. Snow covers the ridge of the ridge.	The **lake in the valley** and the bare land are covered with snow. The mountains are covered with snow covered **lakes and valleys**. The mountain with **lakes** and bare land in the valleys is snowed. Many rivers run through the hill. White snow covers the peaks of the bald mountains.

Fig. 10. Qualitative results of top-5 I2T retrieval for our methods and CLIP*ft* on the RSITMD dataset. Texts marked with green correspond to the ground truth, whereas nonexistent semantic descriptions are marked in red.

Query	Top-5 Retrieval Results Using Our Method	The Ground Truth Retrieval Results
	Beside a bridge full of white iron, **a boat** was moving. Beside a bridge covered with white iron, there was **a moving boat**. On both sides of the dark blue river are white boats. Beside a bridge full of white iron, there was **a moving boat**. Beside a bridge, full of white iron, there was **a movable boat**.	On both sides of the dark blue river are white boats. There are white boats on both sides of the dark blue river. The curved bridge leans against the dark blue river. The winding bridge leans over the dark blue river. A curved bridge tilts a dark blue river.
	The business district is divided into several blocks along the street, each with dense buildings. The business district is divided into pieces by streets, each with dense buildings. The business district is divided into several sections by the streets, each of which has dense buildings. A tightly arranged commercial building. Very orderly commercial buildings.	Each building and each building are tightly side by side. Some buildings surrounded by green trees. The road through the streets. The edge of the road with the many buildings. Each building is tightly crowded between each building.
	Some buildings and **bars Landes** are around the elliptical pond. Several buildings and barley are to the **REDOR** of an oval lake. The lake next to the river is surrounded by a building. The lake next to a river is surrounded by buildings. Several buildings and bare land surround an oval pond.	The lake next to the river is surrounded by a building. On the bank of the river, there is a pond in the park surrounded by residential area. A lake is in the center of the park. This park with a large green pond is surrounded by residential communities. The lake next to a river is surrounded by buildings.
	The **color** of the can is white. Many large storage tanks were built along the road. **Two rows** of storage tanks and several other storage tanks are on bare land near the highway. Many large storage tanks were built along the way. Some tanks are on the side of the road.	**Eight** reservoirs of different sizes are near **Barrellah**. **Eight** tanks of different sizes are located near **Barrelan**. **Eight** tanks of different sizes near **Barrelan**. **Eight** storage tanks of different sizes are near **Barrelan**. There are **eight** tanks of different sizes near **Barriland**.

Fig. 11. Qualitative results of top-5 I2T retrieval using our method on the RSITMD dataset. Green text indicates the ground truth, while red denotes erroneous or nonexistent semantic descriptions.

Query	Top-5 Retrieval Results Using CLIP*ft*	The Ground Truth Retrieval Results
	There is **a boat** under the narrow bridge. There is **a boat** under a narrow bridge. Beside a bridge, full of white iron, there was **a movable boat**. Beside a bridge full of white iron, **a boat** was moving. Beside a bridge full of white iron, there was **a moving boat**.	On both sides of the dark blue river are white boats. There are white boats on both sides of the dark blue river. The curved bridge leans against the dark blue river. The winding bridge leans over the dark blue river. A curved bridge tilts a dark blue river.
	A lot of people are on the street. The business district is divided into several blocks along the street, each with dense buildings. A tightly arranged commercial building. Plenty of skyscrapers are sitting in this busy commercial area. Many cars were parked on the road.	Each building and each building are tightly side by side. Some buildings surrounded by green trees. The road through the streets. The edge of the road with the many buildings. Each building is tightly crowded between each building.
	The **playground** is near some buildings and ponds. A **playground** is near some buildings and a pond. A **playground** is near some buildings and a pond. Some buildings and **bars Landes** are around the elliptical pond. The **playground** is near the building and the pond.	The lake next to the river is surrounded by a building. On the bank of the river, there is a pond in the park surrounded by residential area. A lake is in the center of the park. This park with a large green pond is surrounded by residential communities. The lake next to a river is surrounded by buildings.
	Next to a vertical field of light green, there is a white field. **Plenty of containers** are placed in this bareland. **Two** huge cylindrical **buildings** occupy most of the space. **A white roofed factory** stands on a piece of wasteland. Next to the gray and white building is an open space.	**Eight** reservoirs of different sizes are near **Barrellah**. **Eight** tanks of different sizes are located near **Barrelan**. **Eight** tanks of different sizes near **Barrelan**. **Eight** storage tanks of different sizes are near **Barrelan**. There are **eight** tanks of different sizes near **Barriland**.

Fig. 12. Qualitative results of top-5 I2T retrieval using CLIP*ft* on the RSITMD dataset. Green text indicates the ground truth, while red denotes erroneous or nonexistent semantic descriptions.

descriptions that remain semantically consistent with visual content even when mismatched. Short texts are more prone to retrieval failures than long texts, as the later usually contain distinctive features for correct matching. While our

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model enhances foreground objects, its counting capability and semantic comprehension are constrained by the backbone network, leaving room for further improvement. Furthermore, our method suffer from inappropriate attention focus and diverse descriptive perspectives (e.g. "the color," "two rows"), leading to deviations in retrieval results. In future work, we will explore ranking strategies for coarse texts, object ordering within images, and fine-grained image understanding (e.g., precise counting and small object detection).

V. Conclusion

We proposes FGR-GA, a lightweight MLLM-based framework for remote sensing image–text retrieval, which bridges the cross-modal semantic gap and strengthens the discrimination of similar samples. To extract fine-grained representations, an image feature disentanglement enhancement module and a key term annotation mechanism are introduced to reduce irrelevant information interference for images and text, respectively. Different from instance-level alignment approaches, we develop a group-aware image–text similarity scoring function and a multiview contrastive learning strategy to optimize the shared representation space. Extensive experiments on public benchmarks verify that FGR-GA outperforms state-of-the-art baselines. Future work will explore hard negative sampling, few-shot learning, and text semantic enrichment to further improve retrieval accuracy and robustness.

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